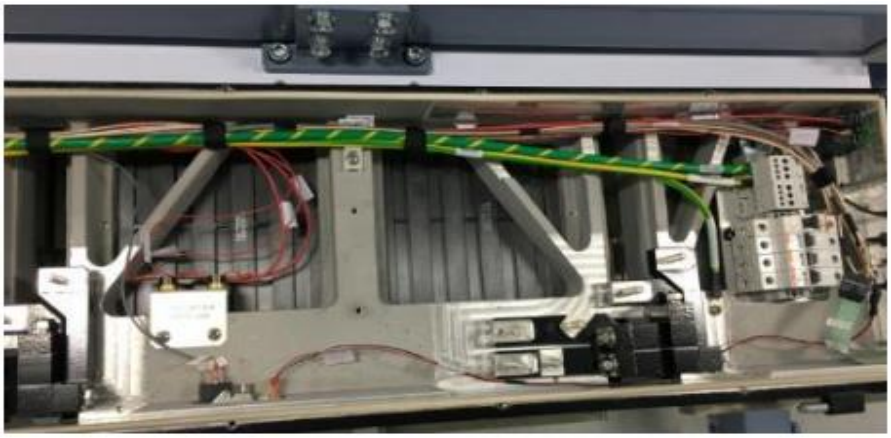
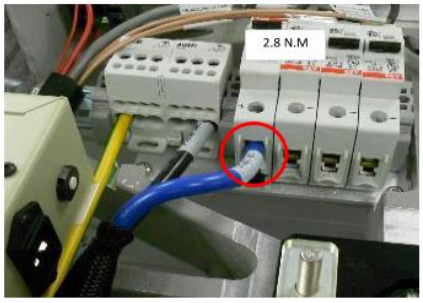
Build Overview:

1. **Circuit Breaker Sub Assembly Build & Install**
2. **Scope Panel Sub Assembly Build & Install**
3. **EMO Button Install**
4. **Cable Install & Routing**
5. **Door Switch Install**
6. **Thermal EMO Switch Install**
7. **Power Strip Install**
8. **Cable Install**
9. **Bus Bar Jumper Install**





CHANGE HISTORY

Date	BPM Rev	Change Details	Updated by
	1	New Issue	JOSEPH AMAD

Prepared By:
JOSEPH R. AMAD
Manufacturing Engineer

Checked By:
IVAN NOWELL PANIQUE
Team Lead

Approved By:
SHERWIN ARIOSTE
Operations Manager

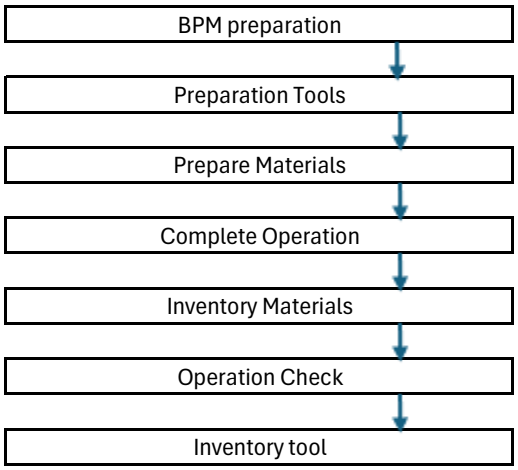
MAN: CERTIFIED TECHNICIANS.

MATERIAL:

NO.	PART NO.	DESCRIPTION	QTY.
1	632-066-06	FORGED OR STAMPED STEEL DIN RAIL 35MM X 7.5MM X 175MM LONG MISC	1
2	632-927-00	ALUMINUM PLATE CIRCUIT BREAKER BRACKET OR MISC ARTICLE	1
3	406-502-06	SCRW PAN HEAD TORX M5X.8P/6MM SS W/NYLOK	2
4	603-393-40	END BRACKET TERMBLOCK 9.5MM GRAY	3
5	653-415-01	CIRCPROTCT MINIATURE BREAKER 40A 3 POLE 500VDC Z CURVE DIN RAIL	1
6	653-525-00	CIRCTPROT MINIATURE CIRCUIT BREAKER -S200-1P-C-20 A	2
7	653-540-00	CIRCTPROT MINIATURE CIRCUIT BREAKER -S200-1P-C-63 A	1
8	357-887-00	CN ONE PHASE POWER DISTRIBUTION BLOCK 1 IN/6 OUT 80A 6OOV DIN RAIL	2
9	601-673-50	CN BUSBAR 4 PINS F/4X1 POLE LENGTH 68MM	1
10	406-502-12	SCRW PAN HEAD TORX M5X.8P/12MM SS W/NYLOK	4
11	631-668-03	ALUMINUM PANEL POWER BRACKET OR MISC ARTICLE	1
12	627-090-00	CABLE UF2 SCOPESYNC PANEL TELECOM WITH CONN	3
13	606-359-35	SCRW FLAT HEAD TORX M4X.7P/35MM SS W/NYLOK	6
14	659-051-00	PLASTIC, LABEL EMERGENCY OFF, EMO PANEL, LABEL	1
15	801-643-01	HDWR SW OPERATOR TU	1
16	801-515-02	SW CONTACT BLOCK 1F	1
17	801-541-02	SW CONTACT BLOCK, NO	1
18	652-712-00	CABLE, UF2 TH INT EMO_LEDS SM CONN, TELECOM, WITH CONN	1
19	629-677-01	CABLE UF2 EMO T/H THERM BRKR LINK TELECOM WITH CONN	2
20	813-929-01	HDW VELCRO STRP 5/8	1
21	652-809-02	CABLE UF2 GROUND JUMPER SP6 TO TERMBLK TELECON WITH CONN	1
22	637-060-05	CABLE, UF2 GND SC TO THTERMBLK, W/CONN	1
23	472-426-12	PAN HEAD CROSS RECESSED SCREW M5X0.8 x 12mm L W/CONE WSHR 5.O.D 1MM	5
24	633-881-02	CABLE, UF2 T/H GROUND WIRE, TELECOM, WITH CONN	1
25	813-929-00	HDW VELCRO STRP 5/8	3
26	938-195-01	CABLE TIE MOUNT 1/4	2
27	801-646-00	SW DOOR INTERLOCK 1	1
28	631-348-00	CABLE UF2 T/H DOOR SWITCH TELECOM WITH CONN	1
29	640-834-01	UOM:EA=mm(drawing defines),PLASTIC, UF2 INST PROTECTIVE	1
30	651-256-00	FORGED OR STAMPED STEEL DIN RAIL 35MM X 7.5MM X 175MM LONG MISC	1
31	470-708-06	ALUMINUM PLATE CIRCUIT BREAKER BRACKET OR MISC ARTICLE	2
32	650-157-00	SCRW PAN HEAD TORX M5X.8P/6MM SS W/NYLOK	1
33	470-626-12	END BRACKET TERMBLOCK 9.5MM GRAY	4
34	812-117-00	CIRCPROTCT MINIATURE BREAKER 40A 3 POLE 500VDC Z CURVE DIN RAIL	2
35	470-353-01	CIRCTPROT MINIATURE CIRCUIT BREAKER -S200-1P-C-20 A	2
36	637-267-00	CIRCTPROT MINIATURE CIRCUIT BREAKER -S200-1P-C-63 A	1
37	634-208-00	CN ONE PHASE POWER DISTRIBUTION BLOCK 1 IN/6 OUT 80A 6OOV DIN RAIL	1
38	202-591-10	CN BUSBAR 4 PINS F/4X1 POLE LENGTH 68MM	2
39	652-436-00	SCRW PAN HEAD TORX M5X.8P/12MM SS W/NYLOK	1
40	675-019-00	ALUMINUM PANEL POWER BRACKET OR MISC ARTICLE	1
41	702-331-02	CABLE UF2 SCOPESYNC PANEL TELECOM WITH CONN	1

METHOD:

Flow chart of operation steps



MACHINE:

TOOL NAME	TORQUE VALUE
FLAT-BLADE SCREWDRIVER	0.7 N.m
PH2 PHILLIPS SCREWDRIVER	1.6 N.m
PH2 PHILLIPS HEAD TOOL	3.3 N.m
16MM SOCKET WRENCH	1.02 N.m
T20 TORX WRENCH	1.6 N.m
EMO NUT TOOL	N/A
SLOTTED SCREWDRIVER	0.7 N.m
16MM SOCKET WRENCH	1.05 N.m(9.3 in-lbs)
FLAT-BLADE SCREWDRIVER	1.6 N.m
PH2 PHILLIPS SCREWDRIVER	0.7 N.m
T20 SCREWDRIVER	3.3 N.m
8MM SLEEVE	0.56 N.m(5 in-lbs)
T15 TORX WRENCH	0.7 N.m
PH2 CROSS CUTTER HEAD	2.8 N.m(28 in-lbs)

MAN: CERTIFIED TECHNICIANS.

MATERIAL:

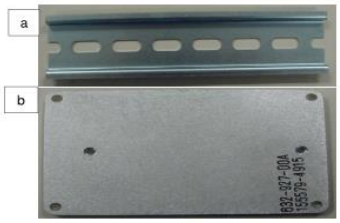
NO.	PART NO.	DESCRIPTION	QTY.
1	631-066-06	FORGED OR STAMPED STEEL DIN RAIL 35MM X 7.5MM X 175MM LONG1 MISC	1
2	632-927-00	ALUMINUM PLATE CIRCUIT BREAKER BRACKET OR MISC ARTICLE	1
3	406-502-06	SCRW PAN HEAD TORX M5X.8P/6MM SS W/NYLOK	2
4	603-393-40	END BRACKET TERMBLOCK 9.5MM GRAY	3
5	653-415-01	CIRCPROTCT MINIATURE BREAKER 40A 3 POLE 500VDC Z CURVE DIN1 RAIL	1
6	653-525-00	CIRCTPROT MINIATURE CIRCUIT BREAKER -S200-1P-C-20 A	2
7	653-540-00	CIRCTPROT MINIATURE CIRCUIT BREAKER -S200-1P-C-63 A	1
8	357-887-00	CN ONE PHASE POWER DISTRIBUTION BLOCK 1 IN/6 OUT 80A 6OOV2 DIN RAIL	2
9	601-673-50	CN BUSBAR 4 PINS F/4X1 POLE LENGTH 68MM	1

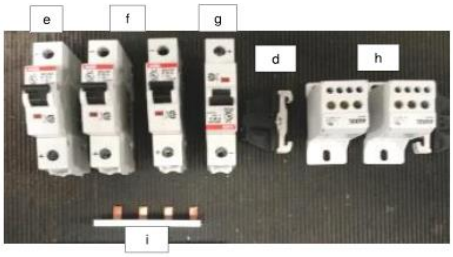
MACHINE: T20 torx wrench, 3.3 N.m

METHOD:

1.1 Take the following materials:

a. Take 1 slide rail (632-066-06).
b. Take 1 bracket (632-927-00).
c. Take two M5 screws (406-502-06).
d. Take three tailstock wiring boards (603-393-40).
e. Take 1 CB 15A (653-415-01).
f. Take 2 CB 20A (653-525-00).
g. Take 1 CB 60A (653-540-00).
h. Take 2 power distribution blocks (357-887-00).
i. Take 1 Busbar (601-673-50).





1.2 Install the slide rail (632-066-06) on the bracket (632-927-00) and fix it with two M5 screws (406-502-06). The tool is a T20 torx wrench. The torque value is 3.3 N·m.

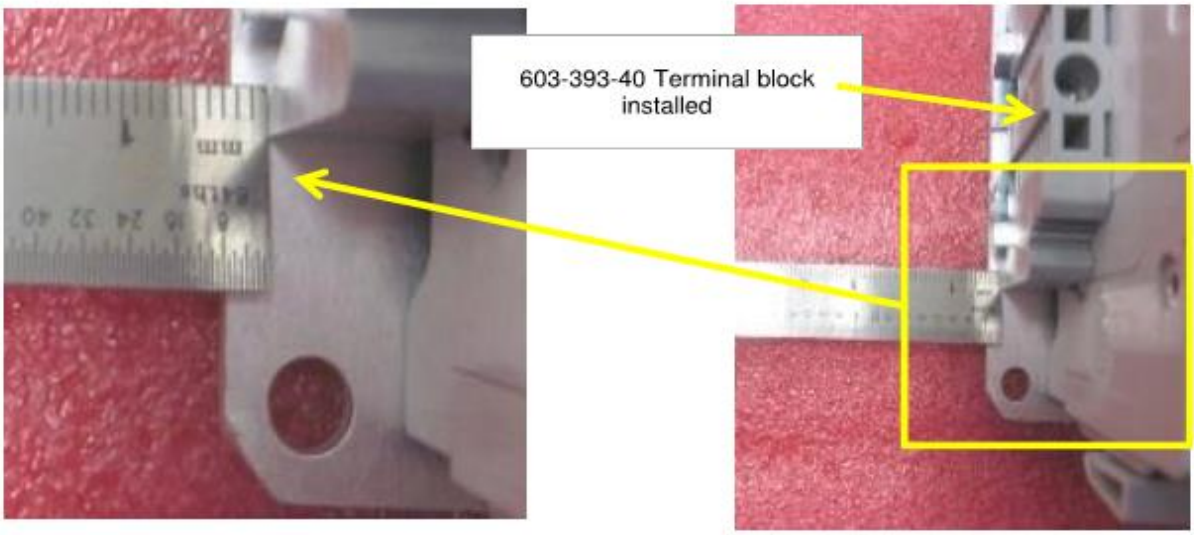
MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: flat-blade screwdriver, 0.7 N.m

METHOD:

1.3 At a distance of 2 mm from the flush end of the bracket and the slide rail, install a tailstock wiring board (603-393-40), the fixing screw tool is a flat-blade screwdriver. The torque value is 0.7 N·m.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

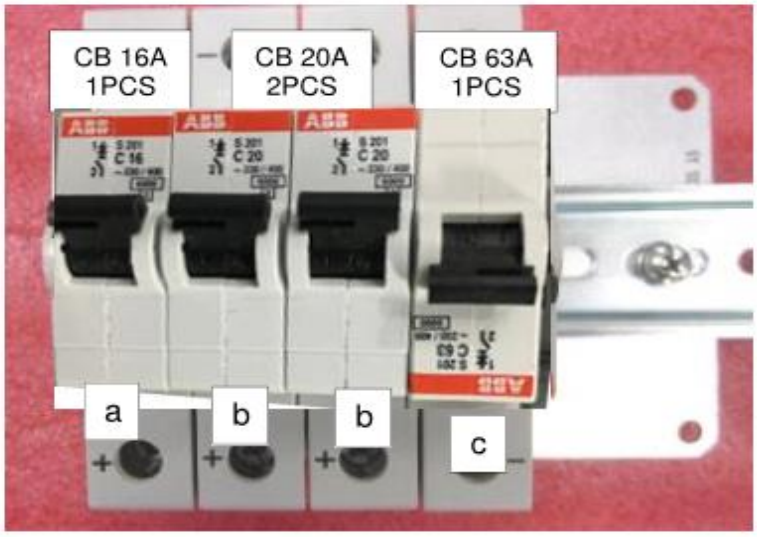
MACHINE: N/A

METHOD:

1.4 Install the circuit breaker on the right side of the tail connector, follow the following assembly sequence as shown in the figure and pay attention to the direction of assembly.

a. Install a CB 16A (653-415-01) on the right end of the tailstock terminal block.
b. Install two CB 20A (653-525-00) on the right side of the circuit breaker 16A.
c. Install a reverse CB 63A (653-540-00) on the right side of the circuit breaker 20A.

Note: Make sure that all circuit breakers are in the "ON" state. As shown in the figure, the red area indicates that the circuit breaker is in the "ON" state.



MAN: CERTIFIED OPERATOR, TECHNICIAN

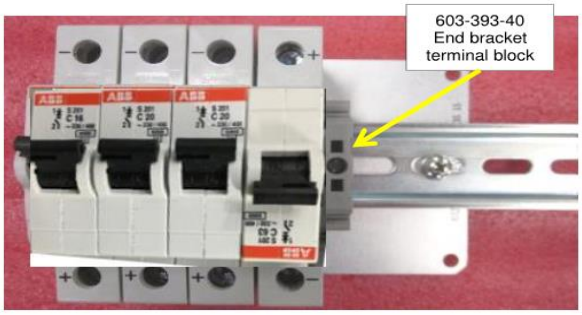
MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
10	406-502-12	SCRW PAN HEAD TORX M5X.8P/12MM SS W/NYLOK	1

MACHINE: flat-blade screwdriver 0.7 N.m

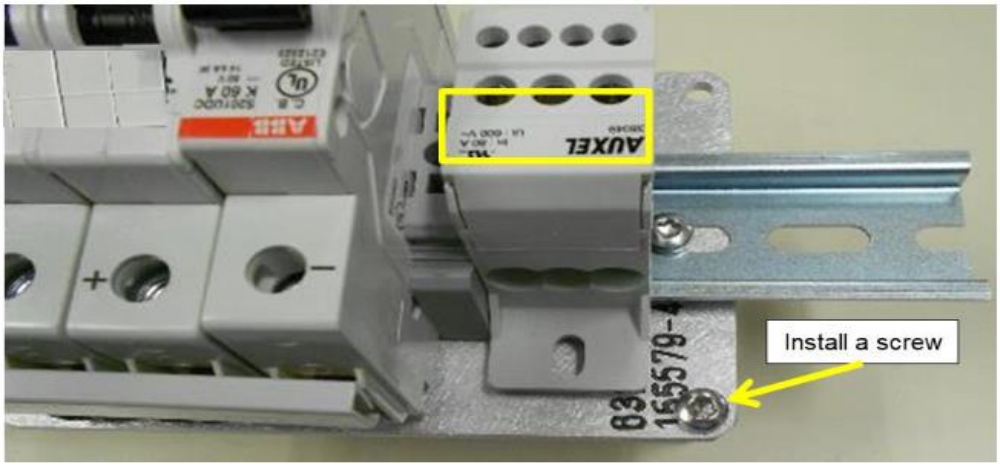
METHOD:

1.5 Install a tailstock wiring board (603-393-40) next to the rightmost circuit breaker.
The fixing screw tool is a flat-blade screwdriver with a torque value of 0.7 N·m.



1.6 Reversely install a power distribution block (357-887-00) on the right side of the tailstock terminal block.
Pay attention to the direction of the power distribution block as shown in the figure.

1.7 Install one screw (406-502-12) as shown. Not tighten yet.



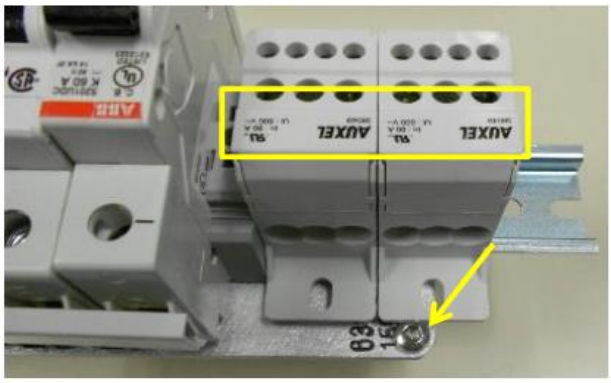
MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

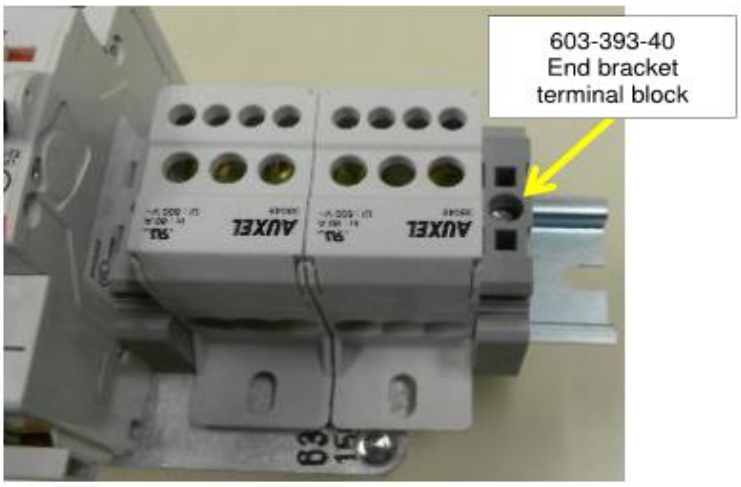
MACHINE: flat-blade screwdriver, 0.7 N.m

METHOD:

1.8 Install a power distribution block (357-887-00) on the right side of this power distribution block in the same direction. Make sure that the screw head is under the Power Distribution Block as shown.



1.9 Install a tailstock wiring board (603-393-40) on the right side of this power distribution block. The fixing screw tool is a flat-blade screwdriver, and the torque value is 0.7 N·m.



MAN: CERTIFIED OPERATOR, TECHNICIAN

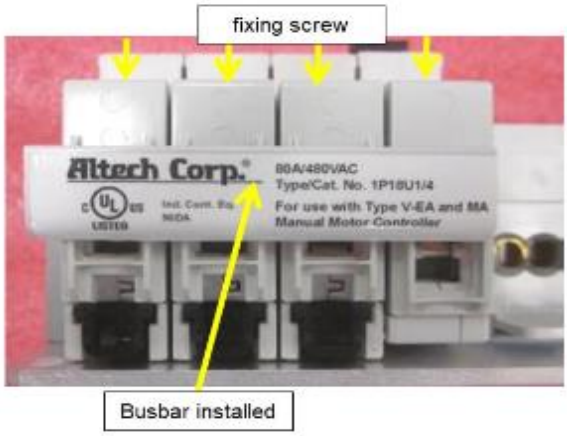
MATERIAL: N/A

MACHINE: PH2 Phillips screwdriver, 2.8 N.m

METHOD:

1.10 Install a Busbar (601-673-50) at the CB port as shown in the figure, and use a PH2 Phillips screwdriver to fix the screw to 2.8 N·m. A total of four locations.

Install busbar at CB ports as shown



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

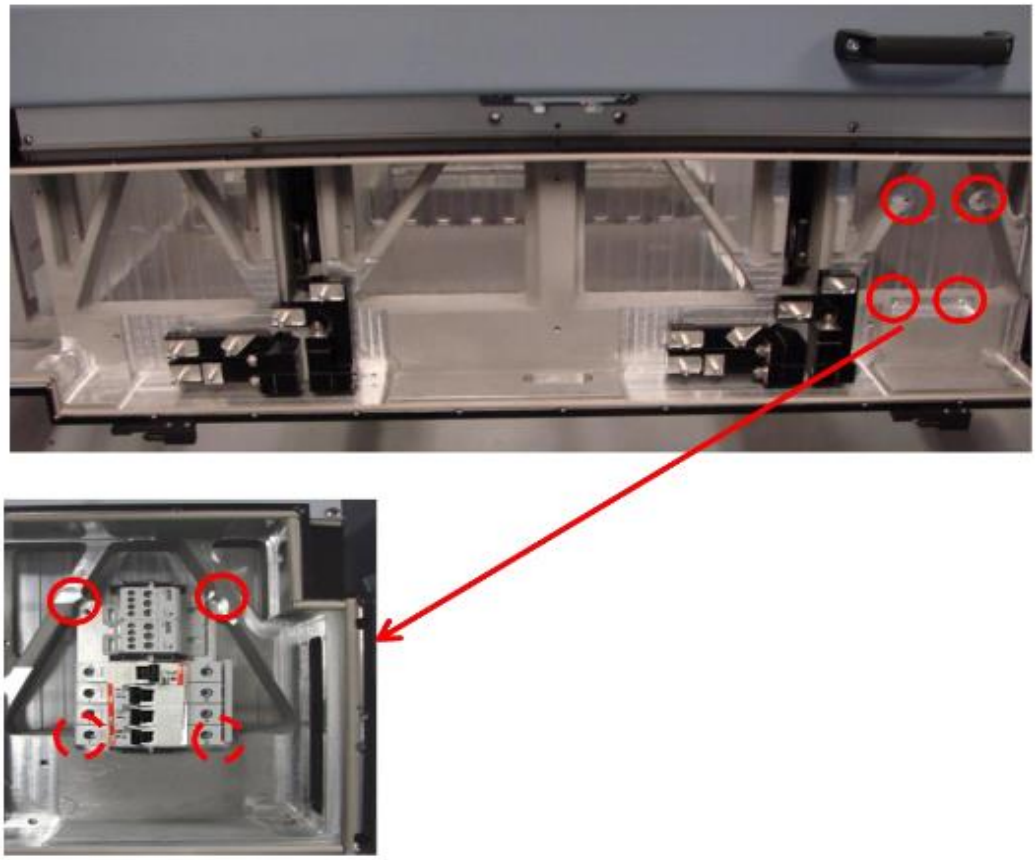
ITEM	PART NUMBER	DESCRIPTION	QTY
11	406-502-12	SCRW PAN HEAD TORX M5X.8P/12MM SS W/NYLOK	3

MACHINE: T20, 3.3 N.m

METHOD:

1.12 Rotate the T/H direction as shown in the figure.

1.13 Install the circuit breaker components as shown in the figure, please pay attention to the direction when installing. One of the screws is assembled before, install 3 in another position M5 screws (406-502-12). Use the bit T20 tool to fix the screw with a torque value of 3.3 N·m.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
11	631-668-03	ALUMINUM PANEL POWER BRACKET OR MISC ARTICLE	1
12	627-090-00	CABLE UF2 SCOPESYNC PANEL TELECOM WITH CONN	3

MACHINE: Marker

METHOD:

2.1 Take the following materials as shown:

a. Take 1 panel (631-668-03).

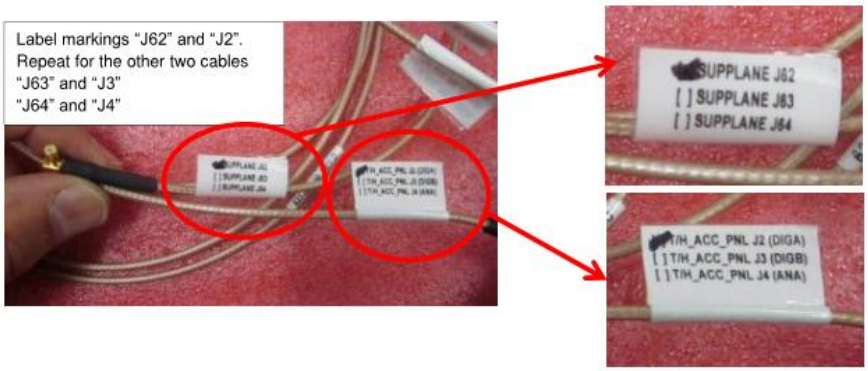
b. Take 3 wires (627-090-00).

2.2 Mark the labels of the lines in the following order as shown:

a. Mark "J4" on one end of the line and "J64" on the other end.

b. Mark "J3" on one end of the line and "J63" on the other end.

c. Mark "J2" at one end of the line and "J62" at the other end.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

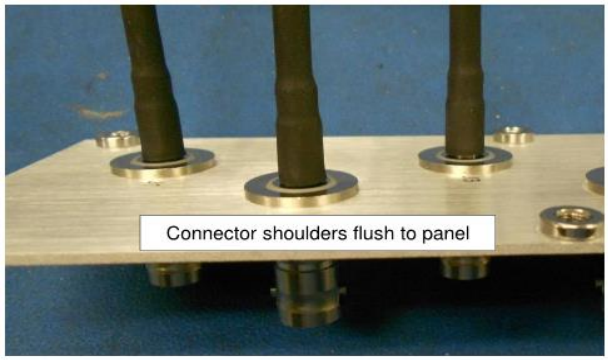
MACHINE: 16mm socket wrench, 1.02 N.m

METHOD:

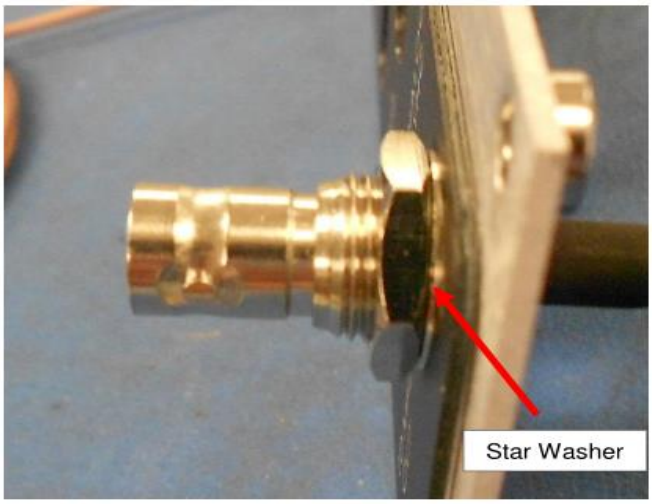
2.3 Remove the nut and star washer from the connector.

2.4 Insert 3 wires (627-090-00) into the panel (631-668-03) according to the label.

Note: When inserting the wire, make sure that the connector shoulders is flush with the panel surface.



2.5 Put a star washer between the nut and the panel, use a 16mm socket wrench to fix the connector, and the torque value is 1.02 N.m. (9 in-lbs). There are three connectors in total.



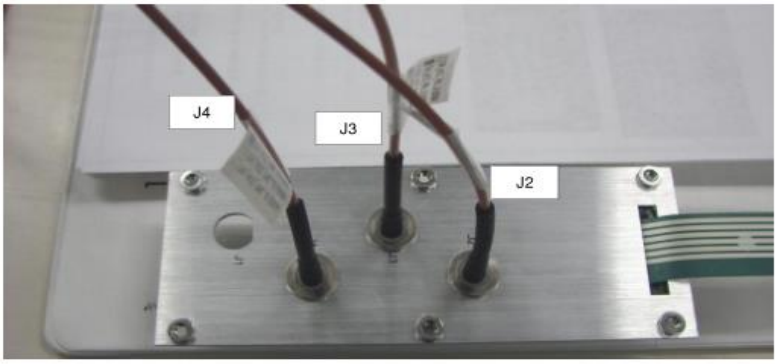
MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

2.6 The panel is assembled as shown.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

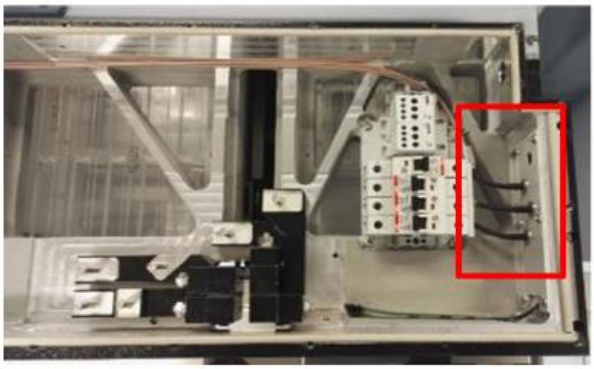
ITEM	PART NUMBER	DESCRIPTION	QTY
14	606-359-35	SCRW FLAT HEAD TORX M4X.7P/35MM SS W/NYLOK	6

MACHINE: T20 Torx wrench, 1.6 N.m

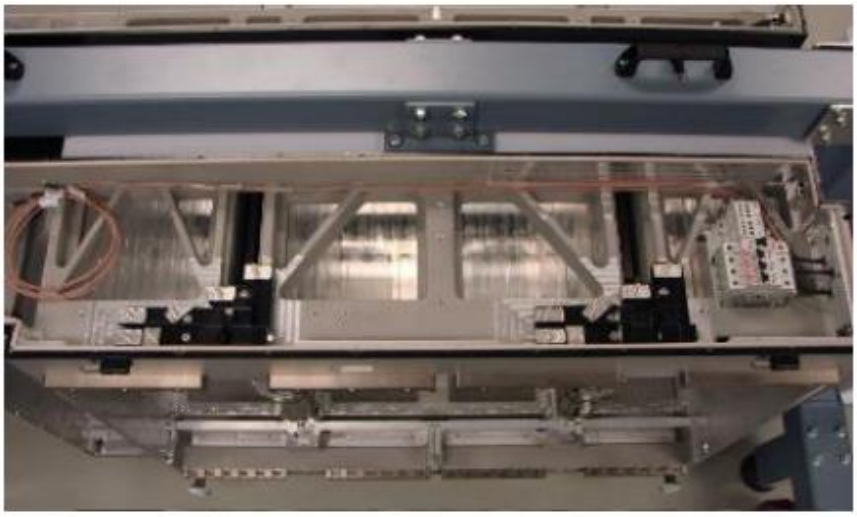
METHOD:

3.1 Install the panel to the T/H frame as shown in the figure, install 6pcs M4 screws (606-359-35), and fix it with the T20 Torx wrench. The torque value is 1.6 N·m.

3.2 Carefully route the wire around the circuit breaker as shown in the figure, and place the wire along the T/H frame without bending the wire



3.3 Carefully place the line along the leftmost side of the T/H frame as shown in the picture, and make a circle.



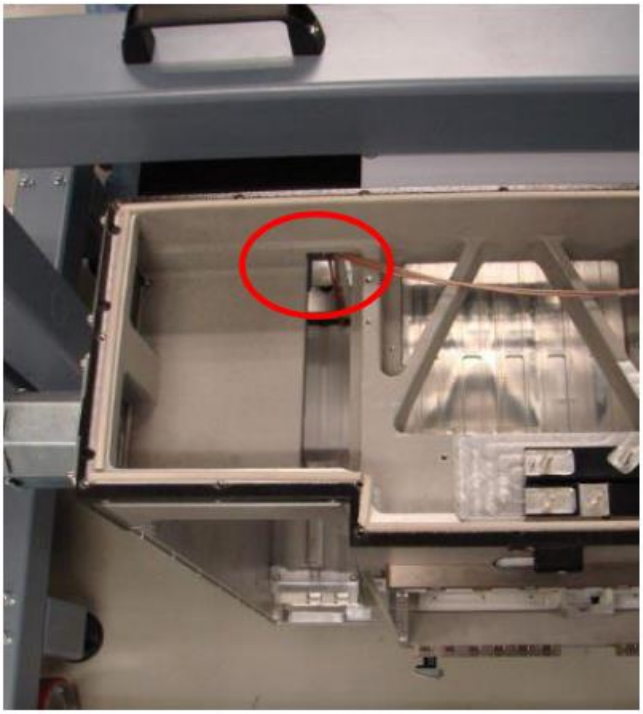
MAN: CERTIFIED OPERATOR, TECHNICIAN

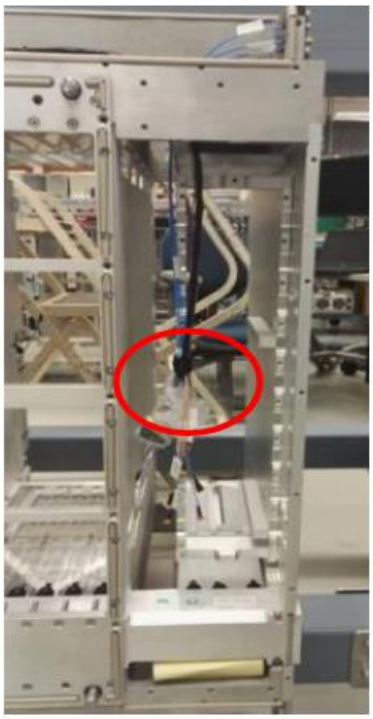
MATERIAL: N/A

MACHINE: N/A

METHOD:

3.4 Pass the wire through the opening of the T/H frame and face the backplane as shown in the figure, and tie the wire harness to the original place with the existing black wire tie.





MAN: CERTIFIED TECHNICIANS.

MATERIAL:

NO.	PART NO.	DESCRIPTION	QTY.
16	631-066-06	FORGED OR STAMPED STEEL DIN RAIL 35MM X 7.5MM X 175MM LONG1 MISC	1
17	632-927-00	ALUMINUM PLATE CIRCUIT BREAKER BRACKET OR MISC ARTICLE	1
18	406-502-06	SCRW PAN HEAD TORX M5X.8P/6MM SS W/NYLOK	2
19	603-393-40	END BRACKET TERMBLOCK 9.5MM GRAY	3
20	653-415-01	CIRCPROTCT MINIATURE BREAKER 40A 3 POLE 500VDC Z CURVE DIN1 RAIL	1

MACHINE: EMO nut tool

METHOD:

4.1 Take the following materials as shown:

a. Take an operation switch (801-643-01).

b. Take a switch contact block (801-515-02).

c. Take a switch contact block (801-541-02).

d. Take a LED EMO cable (652-712-00).

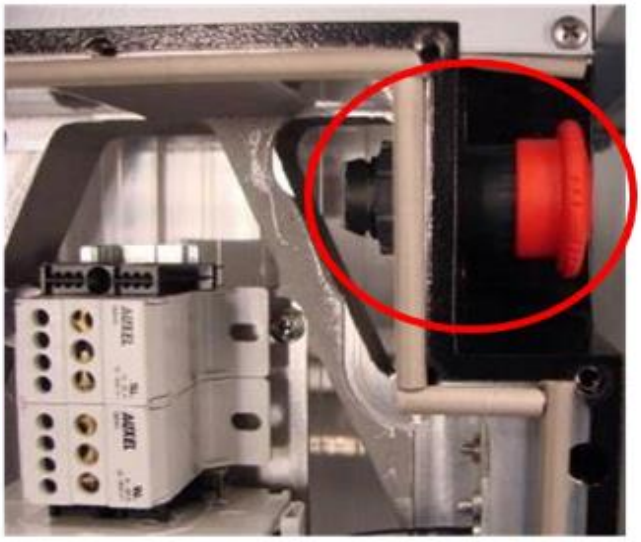
f. Take 2 EMO cables (629-677-01).

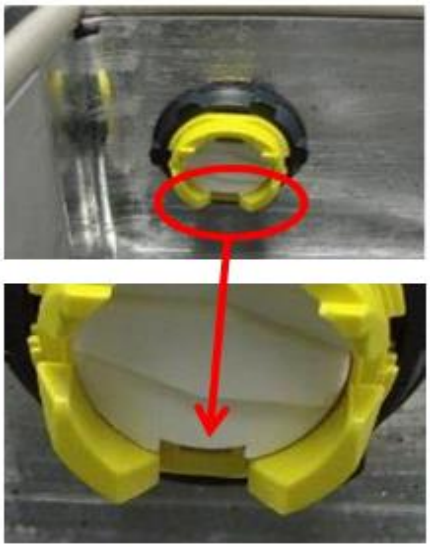
4.2 Install the switch knob to the T/H frame as shown in the figure.

a. Remove the nut on the switch knob.

b. Install the switch knob into the T/H frame hole near the circuit breaker, and fix it with the nut removed before. Use EMO nut tool to fix the nut. The torque value is 2.0 N.m.(17.7 in-lbs.)

c. Note that the notch of the EMO knob switch faces downward as shown.





MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

4.3 Insert the switch contact block (801-515-02) into another switch contact block (801-541-02) as shown.

4.4 Take 2 EMO wires (629-677-01) as shown in the figure and mark them according to the following requirements.

a. One cable (629-677-01) is marked "TH_EMO-T1" at one end. The other end is marked "TH_EMO – TS1B".

b. One cable (629-677-01) is marked "TH_EMO – T2" at one end. The other end is marked "TH_EMO – TS2B".

Label markings “TH_EMO – T2” and TH_EMO –TS2B” Repeat for the other cable “TH_EMO – T1” and TH_EMO –TS1B

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

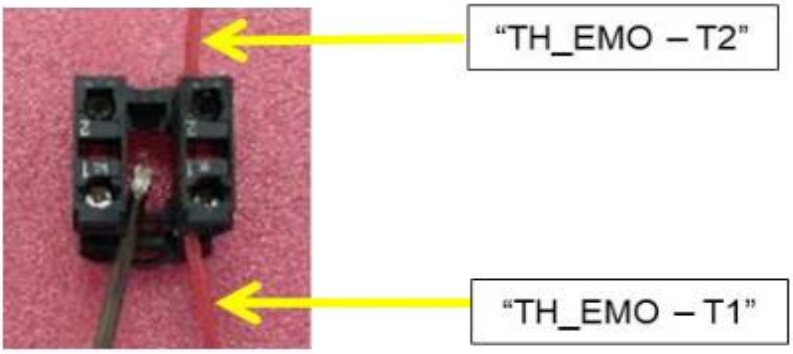
MACHINE: flat-blade screwdriver 0.7 N.m

METHOD:

4.5 Install the wire according to the label as shown in the figure.

a. Install the line "TH_EMO – T1" at position 1 and fix it with a flat-blade screwdriver with a torque value of 0.7 N·m.

b. Install the line "TH_EMO – T2" in position 2 and fix it with a flat-blade screwdriver. The torque value is 0.7 N·m.



MAN: CERTIFIED OPERATOR, TECHNICIAN

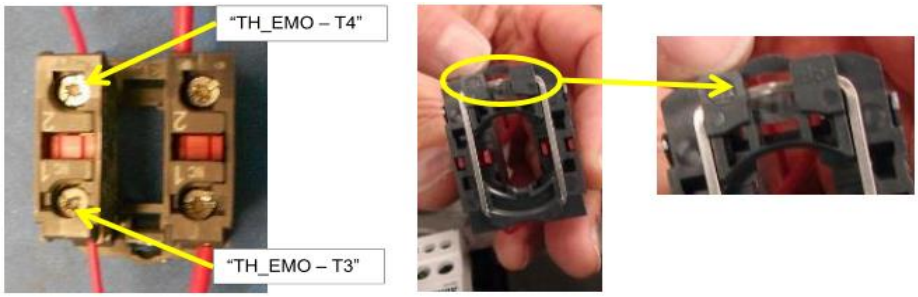
MATERIAL: N/A

MACHINE: Flat screwdriver 0.7 N.m

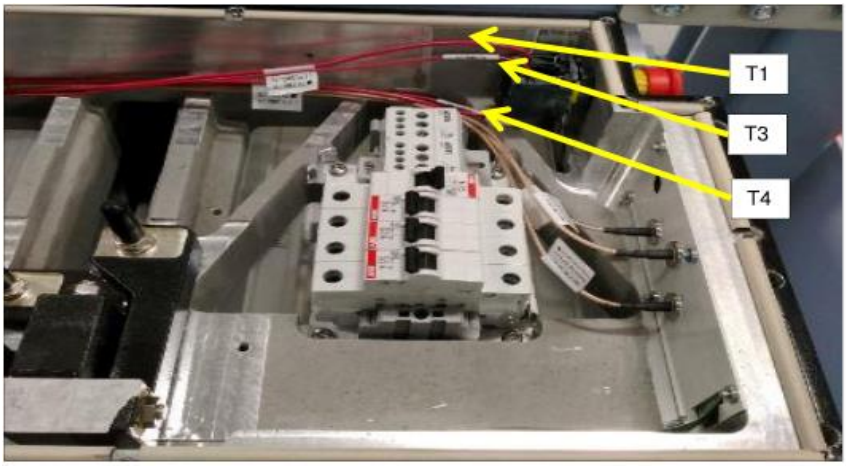
METHOD:

5.1 Install an LED EMO cable (652-712-00) as shown.

a. Install the switch terminals "1" and "2" according to the line labels "TH_EMO-T3" and "TH_EMO-T4", use Flat screwdriver, the torque value is 0.7 N·m.



5.2 Install the switch contact block to the EMO switch on the T/H as shown in the figure. Make sure that the bridge is up when installing and the switch contact block is fully engaged with the EMO switch.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
21	813-929-01	HDW VELCRO STRP 5/8	1

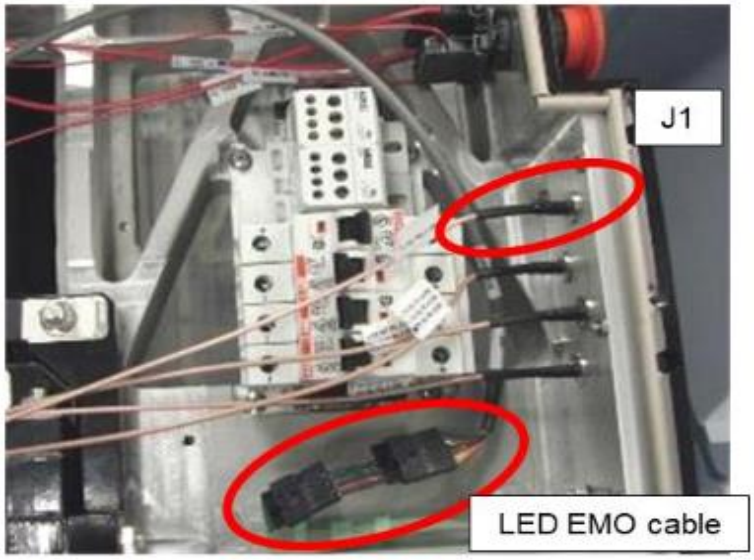
MACHINE: 16mm socket wrench 1.05 N.m(9.3 in-lbs)

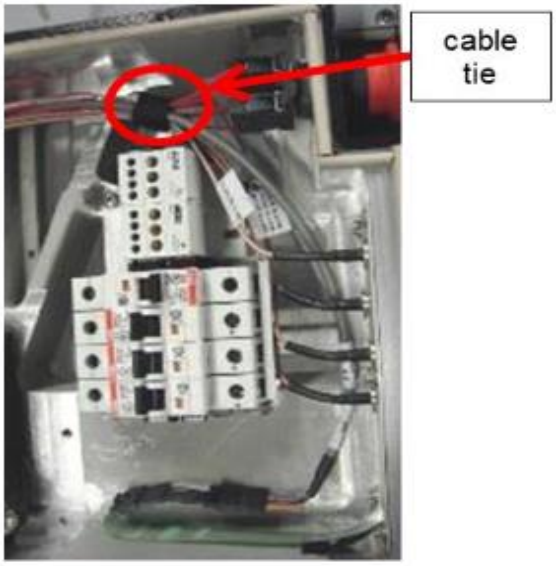
METHOD:

5.3 Connect the end of the label “T/H LEDPNL CN1” on the LED EMO line to the harness connector on the upper panel as shown in the figure.

5.4 Install the 10 GHZ Scopesync cable end label marked "T/H_ACC_PNL J1 to J1" as shown in the figure.
Use a 16mm socket wrench at J1 of the panel. Secure the nut to 1.05 N.m.(9.3 in-lbs.)

5.5 Route the wire around the circuit breaker as shown in the figure and place it along the T/H frame.
Use a cable tie (813-929-01) to tighten the thread.





MAN: CERTIFIED OPERATOR, TECHNICIAN

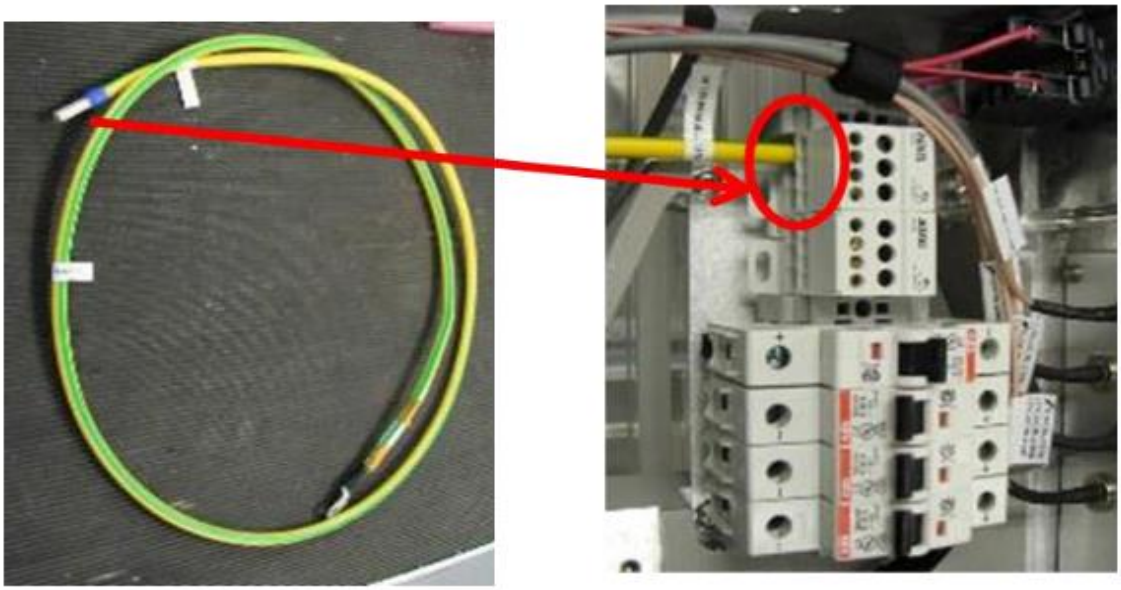
MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
22	652-809-02	CABLE UF2 GROUND JUMPER SP6 TO TERMBLK TELECOM WITH CONN1	1

MACHINE: PH2, 1.6 N.m

METHOD:

5.6 Install a ground wire (652-809-02) on the first space above the power distribution block as shown in the figure.
Use a PH2 to secure the torque value is 1.6 N·m.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

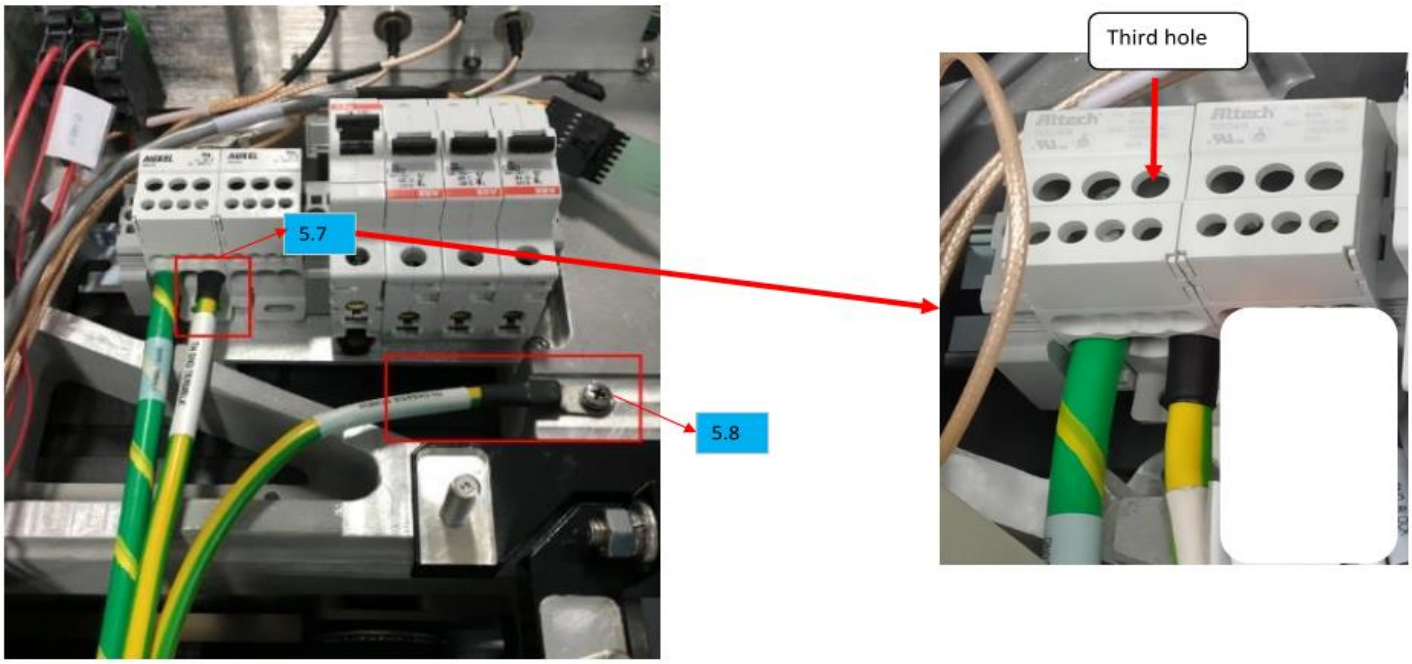
ITEM	PART NUMBER	DESCRIPTION	QTY
23	689-743-00	CABLE, UF2 GND THTERMBLK AND TH CHASIS, W/CONN	1
24	472-426-12	PAN HEAD CROSS RECESSED SCREW M5X0.8 x 12MM L W/CONE WASHER. O.D. 1 11MM	3

MACHINE: PH2, Torque 1.6 N.m and PH2 with Torque 3.3 N.m

METHOD:

5.7 Install a ground wire (689-743-00) Label TH GND TERMBLK in the third hole of the power distribution block as shown in the figure. Use a PH2 to secure the torque value is 1.6 N·m.

5.8 Use an M5 screw (472-426-12) to install a ground wire (689-743-00) Label TH CHASIS SCRW on the hole on the cabinet next to the switch as shown in the figure.



MAN: CERTIFIED OPERATOR, TECHNICIAN

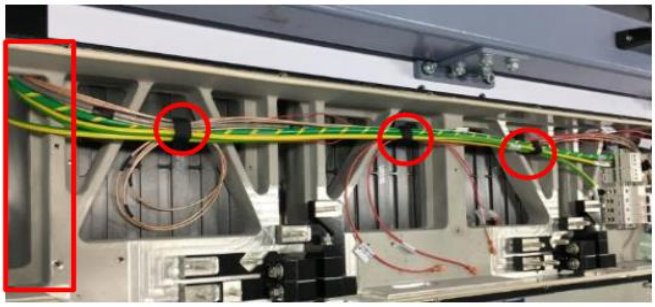
MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
26	813-929-00	HDW VELCRO STRP 5/8	3
27	938-195-01	CABLE TIE MOUNT 1/4	2
28	472-426-12	PAN HEAD CROSS RECESSED SCREW M5X0.8 x 12MM L W/CONE WASHER.2 O.D.11 MM	2

MACHINE: PH2 with Torque 3.3 N.m

METHOD:

5.9 Place the wire harness along the T/H frame as shown in the figure, and use three cable ties (813-929-00) to tighten the wire harness. Thread the EMO wire and ground wire.
At the opening of the T/H frame, the line is perpendicular to the backplane.



5.10 Fix the two wire ties (938-195-01) with two M5 screws (472-426-12) as shown in the figure, the tool PH2 cross bit, the torque value is 3.3 N·m.

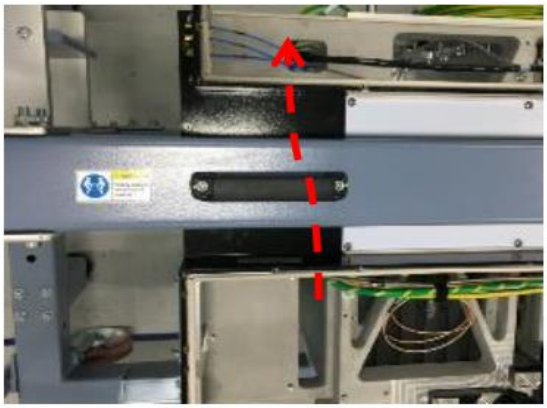
MAN: CERTIFIED OPERATOR, TECHNICIAN

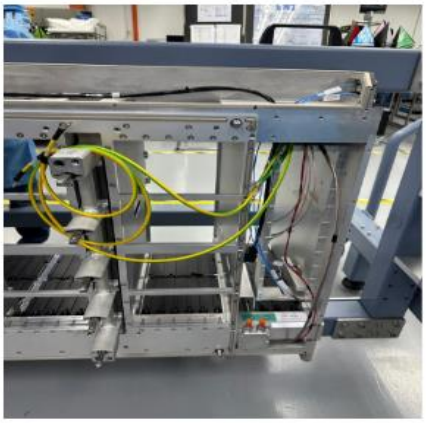
MATERIAL: N/A

MACHINE: N/A

METHOD:

5.11 Wind the ground wire and EMO wire to the back panel as shown in the figure, pass through the uppermost channel, cross the T/H frame, and pass out of the channel.
Temporarily wrap two long ground wires around the IPD and fix them on the frame.
The short ground wire will be installed on the SP4 board in the subsequent steps.





MAN: CERTIFIED OPERATOR, TECHNICIAN

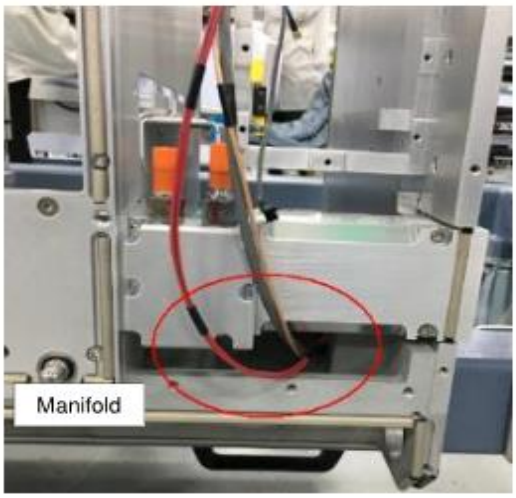
MATERIAL: N/A

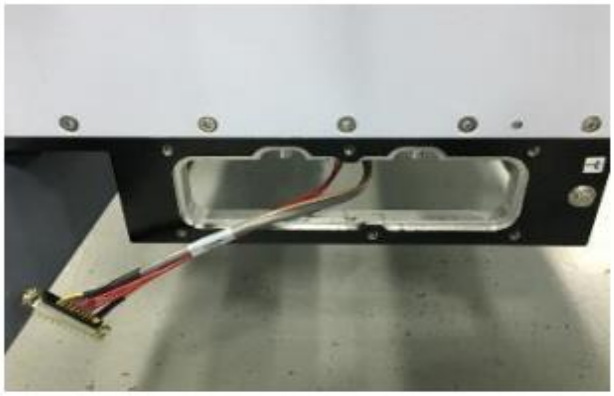
MACHINE: N/A

METHOD:

5.12 Continue to by pass the Manifold and pass the EMO through the channel.

Pass the EMO cable through the dog house channel interface and temporarily hang it on the frame.





MAN: CERTIFIED TECHNICIANS.

MATERIAL:

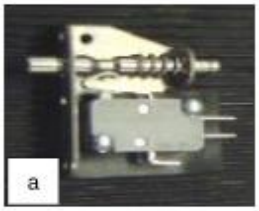
NO.	PART NO.	DESCRIPTION	QTY.
29	801-646-00	SW DOOR INTERLOCK 1	1
30	631-348-00	CABLE UF2 T/H DOOR SWITCH TELECOM WITH CONN	1
31	651-256-00	ALUMINUM, INTERLOCK SWITCH, BRACKET OR MISC ARTICLE	1
32	470-708-06	SCRW W/WSHR 6-32X3/8" STL/	2
33	650-157-00	CABLE UF2 LED TH POWER JUMPER TELECOM WITH CONN	1
34	702-331-02	RED OPTO.LED.	1
35	675-019-00	MECHANICAL, TH DOOR STOP PLATE, FOR SEMI TEST	1

MACHINE: N/A

METHOD:

6.1 Take the following materials as shown:

a. Take 1 door lock switch (801-646-00).
b. Take 1 door lock switch wire (631-348-00).
c. Take 1 bracket (651-256-00).
d. Take 2 6-32 screws (470-708-06).
e. Take 1 LED wire (650-157-00).
f. Take 1 LED light (702-331-02).
g. Take 1 piece of iron (675-019-00).



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: PH2 Phillips screwdriver 0.7 N.m

METHOD:

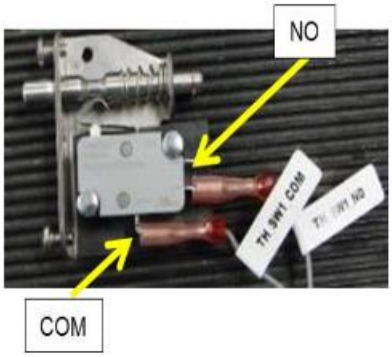
6.2 Install the door lock wire (631-348-00) to the door lock switch (801-646-00) according to the label as shown:

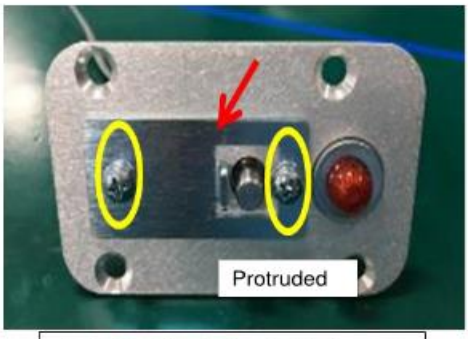
a. Install the line "TH_SW1_COM" to the COM terminal of the door lock switch.

b. Install the line "TH_SW1_NO" to the NO end of the door lock switch.

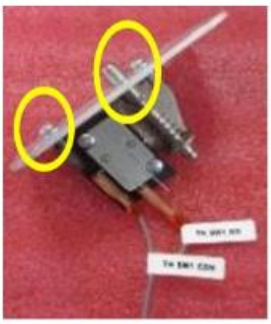
6.3 As shown in the figure, install the door lock switch and a piece of iron (675-019-00) to the bracket (651-256-00) in the direction shown in the figure. Fix it with 2x 6-32 screws (470-708-06), the tool is a PH2 Phillips screwdriver, and the torque value is 0.7 N·m.

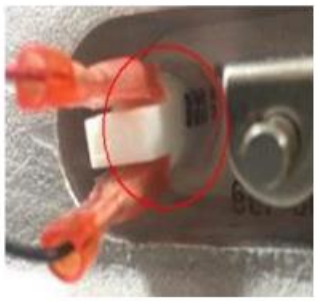
6.4 Insert the LED light (702-331-02) into the bracket as shown in the figure, and pay attention to the direction that the LED light faces the outside of the bracket; And insert the LED wire (650-157-00) into the insert on the back of the LED light according to the corresponding'+''-'.





Installing door lock switch to bracket





MAN: CERTIFIED OPERATOR, TECHNICIAN

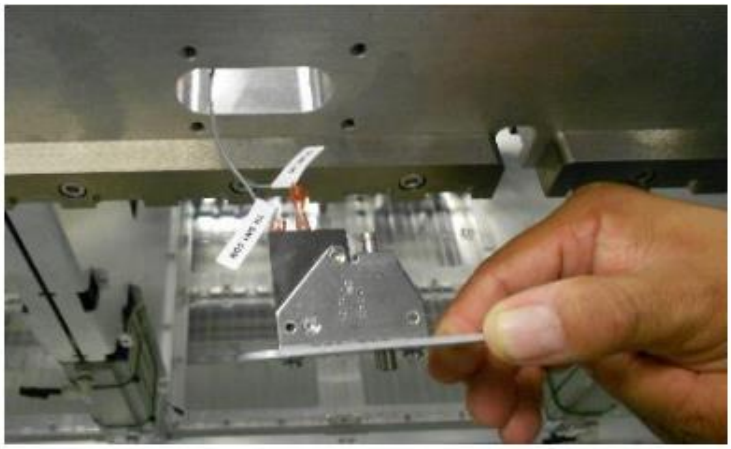
MATERIAL:

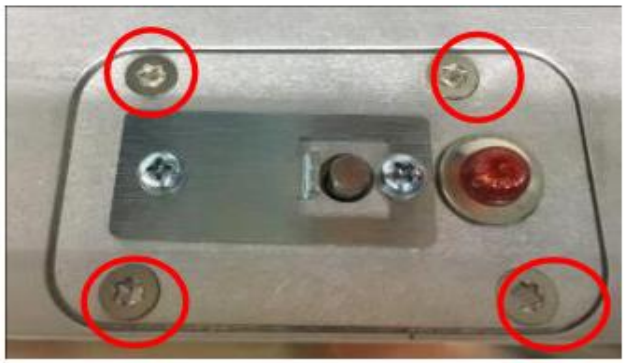
ITEM	PART NUMBER	DESCRIPTION	QTY
34	470-626-12	SCRW FLAT HEAD TORX M5X.8P/12MM SS W/NYLOK	4

MACHINE: T20 screwdriver 3.3 N.m

METHOD:

6.5 Pass the door lock switch wire through the small hole as shown, and fix the door lock to the T/H frame with 4pcs M5 screws (470-626-12). The tool is a T20 screwdriver. The torque value is 3.3 N·m.





MAN: CERTIFIED OPERATOR, TECHNICIAN

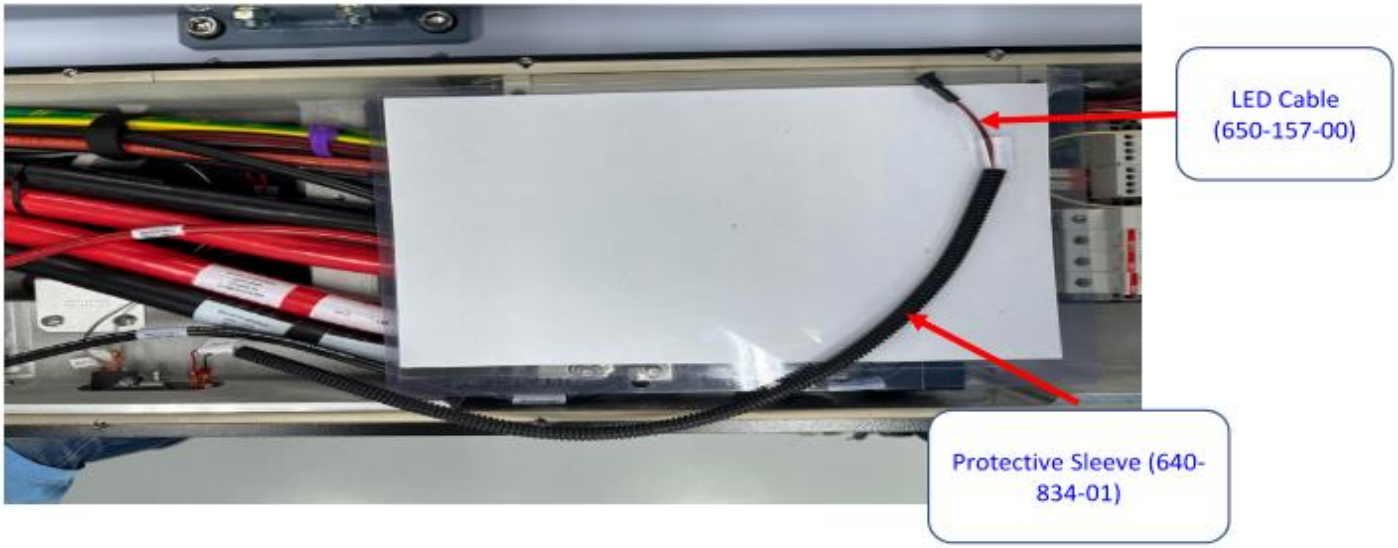
MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
1	640-834-01	UOM:EA=mm(drawing defines),PLASTIC, UF2 INST PROTECTIVE	1

MACHINE: N/A

METHOD:

6.6 Install 1pcs (640-834-01) to led cable (650-157-00) and put it as shown in diagram



MAN: CERTIFIED OPERATOR, TECHNICIAN

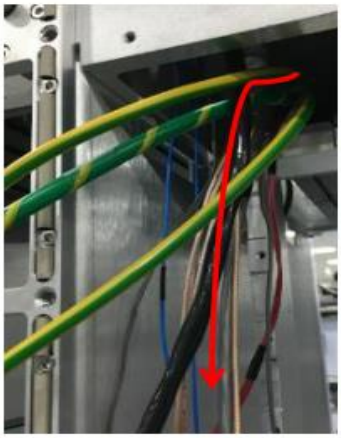
MATERIAL: N/A

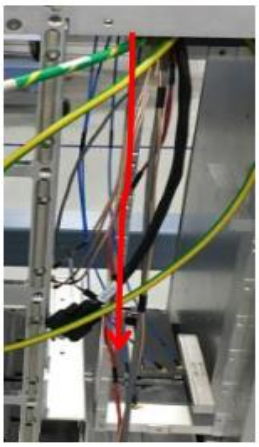
MACHINE: N/A

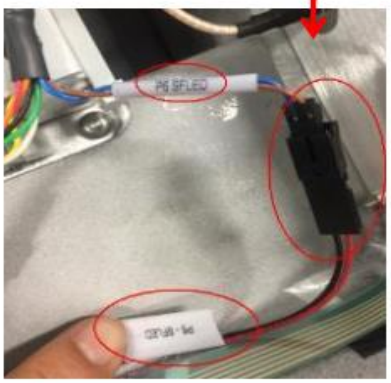
METHOD:

6.7 Pass the switch wire into the small hole along the T/H frame as shown in the figure, go through the channel to the back panel surface, and use the existing cable tie to tie the wire tightly.

6.8 Connect the 650-157-00 LED wire connector 'P6-SFLED' to the 'P6 SFLED' connector of 652-712-00 as shown in the figure.







Connect connector LED wire 'P6-SFLED' to the 'P6 SFLED' connector

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
35	812-117-00	SW THERMO 70C OPEN	2
36	470-353-01	NUT KEPS 6-32FL RIM CONL W	2
37	637-267-00	ALUMINUM PLATE THERMAL SWITCH BRACKET OR MISC ARTICLE	1

MACHINE: N/A

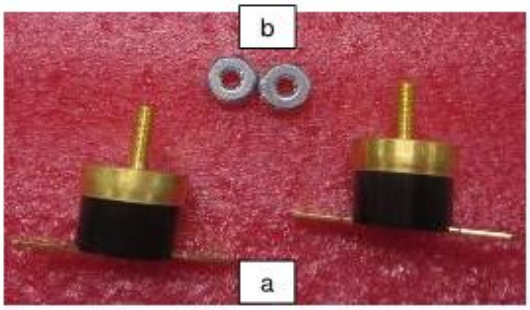
METHOD:

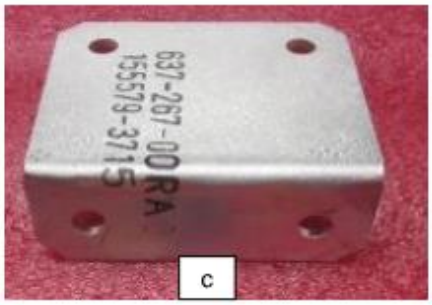
7.1 Take the following materials as shown:

a. Take 2 temperature control switches (812-117-00).

b. Take 2 6-32 nuts (470-353-01).

c. Take 1 bracket (637-267-00).





7.2 Install 2 temperature control switches (812-117-00) on the bracket (637-267-00) as shown in the figure, and fix them with 2 nuts (470-353-01).

The direction of the temperature control switch is shown in the figure. Manually tighten the nut but do not tighten it.

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

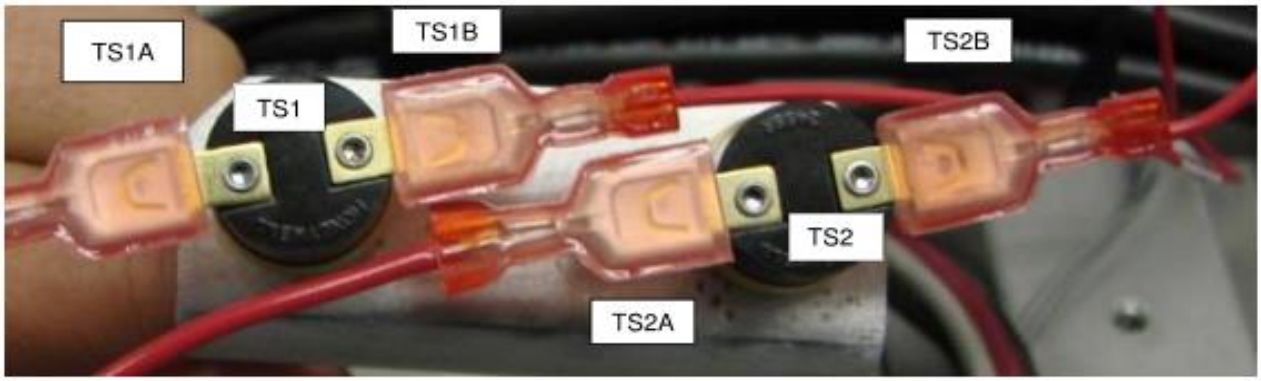
MACHINE: 8mm sleeve 0.56 N.m (5 in-lbs)

METHOD:

7.3 Install the EMO line on the temperature control switch as shown:

a. Insert the line label "TH_EMO-TS1A" into TS1A, and the line label "TH_EMO-TS2A" into TS2A.

b. Insert the line label "TH_EMO-TS1B" into TS1B, and the line label "TH_EMO-TS2B" into TS2B.



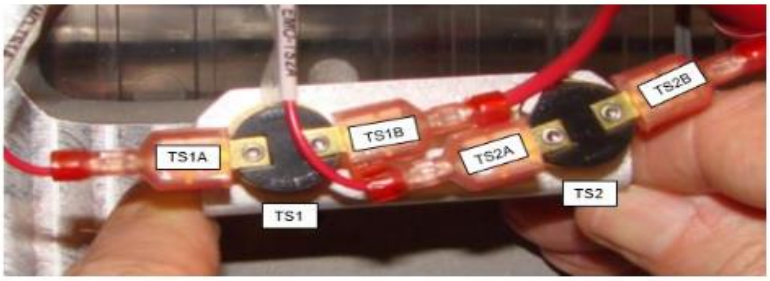
MAN: CERTIFIED OPERATOR, TECHNICIAN

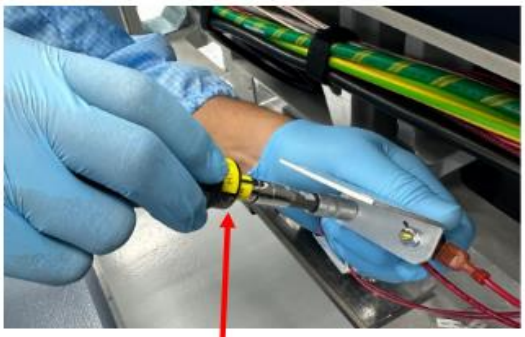
MATERIAL: N/A

MACHINE: 8mm sleeve 0.56 N.m (5 in-lbs)

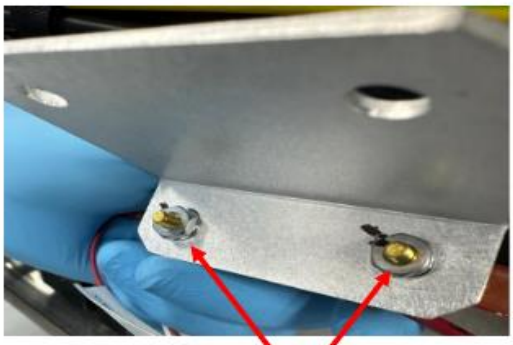
METHOD:

7.3 Rotate the direction of the temperature control switch as shown, and place the two wires in the middle as close as possible. Use a tool 8mm sleeve to fix the torque value to 0.56 N.m. (5 in-lbs). There are two places. Apply torque marking at 2 pcs nut after torque





Torque 0.56Nm with 8mm Socket



Apply **Torque Marking** after done torque

MAN: CERTIFIED OPERATOR, TECHNICIAN

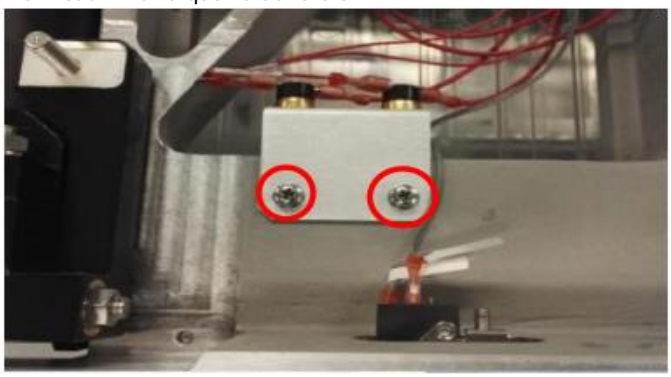
MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
38	472-426-12	PAN HEAD CROSS RECESSED SCREW M5X0.8 x 12MM L W/CONE WASHER. O.D. 11MM	2

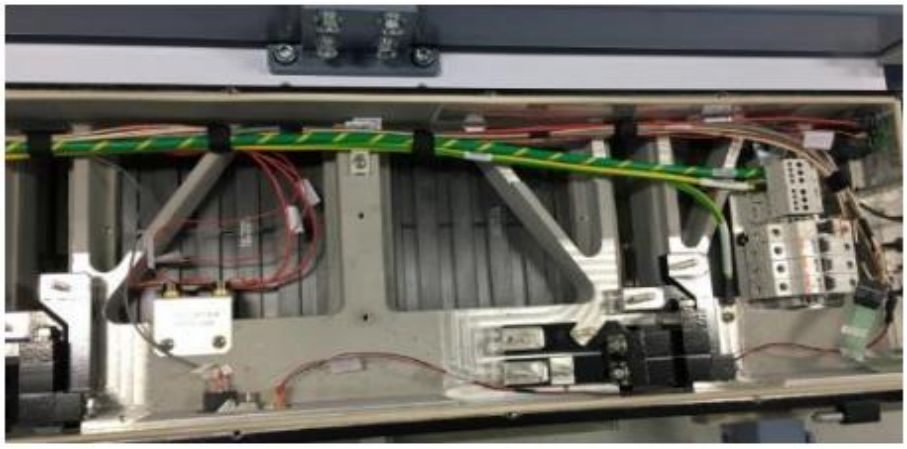
MACHINE: PH2, 1.6 N.m

METHOD:

7.3 Install the temperature control switch on the T/H frame as shown in the figure, fix it with 2pcs M5 screws (472-426-12). The tool is PH2 cross cutter head. The torque value is 3.3 N·m.



7.4 As shown in the figure after installation.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

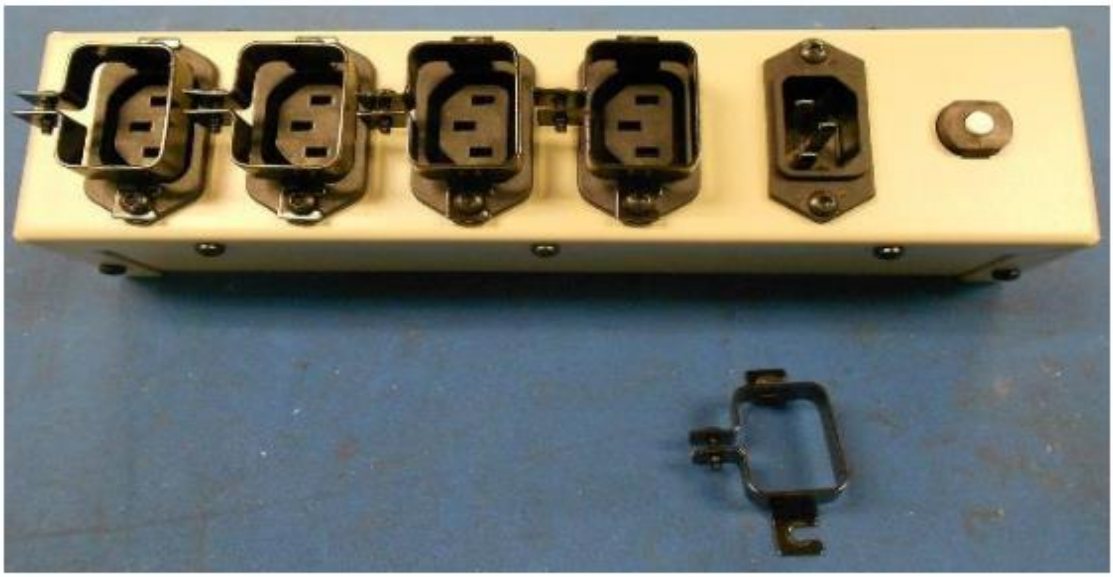
ITEM	PART NUMBER	DESCRIPTION	QTY
39	634-208-00	POWER STRIP 4 POSITIONS W CONN LOCKS 10A 125-240VAC	1

MACHINE: T15 torx wrench 0.7 N.m

METHOD:

8.1 Take the power strip (634-208-00) as shown in the figure and use the tool T15 torx wrench to loosen the 2 screws of the AC cable clamp.

8.2 Remove the clamp and re-use a T15 torx wrench to fix the screw. The torque value is 0.7 N·m.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

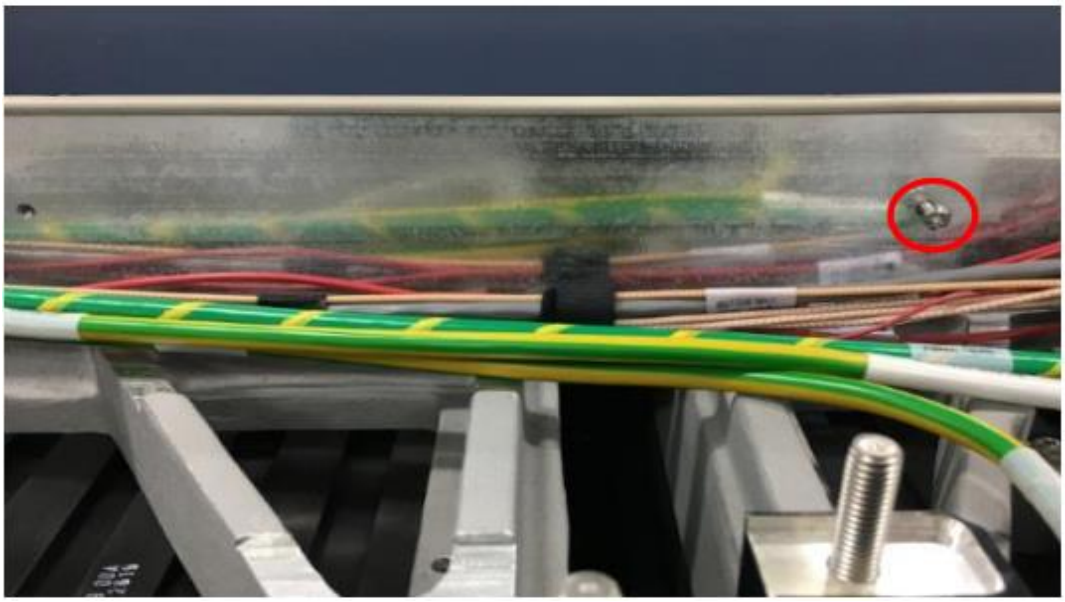
ITEM	PART NUMBER	DESCRIPTION	QTY
40	202-591-10	SCRW PAN HEAD TORX M4X.7P/10 MM SS W/NYLOK	1

MACHINE: N/A

METHOD:

8.3 Before installing the patch panel as shown in the figure, ensure that the EMO cable and the ground cable are placed flat on the T/H frame and do not cross each other.

8.4 Install a screw (202-591-10) in the hole on the right, do not tighten it, the patch panel will be installed here.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

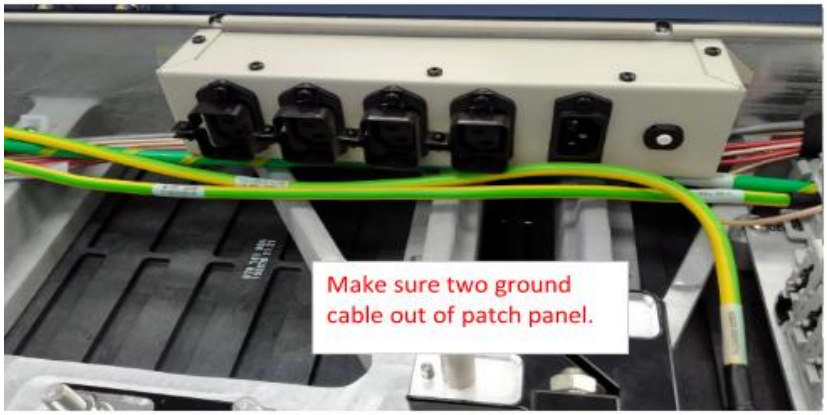
ITEM	PART NUMBER	DESCRIPTION	QTY
41	202-591-10	SCRW PAN HEAD TORX M4X.7P/10 MM SS W/NYLOK	1

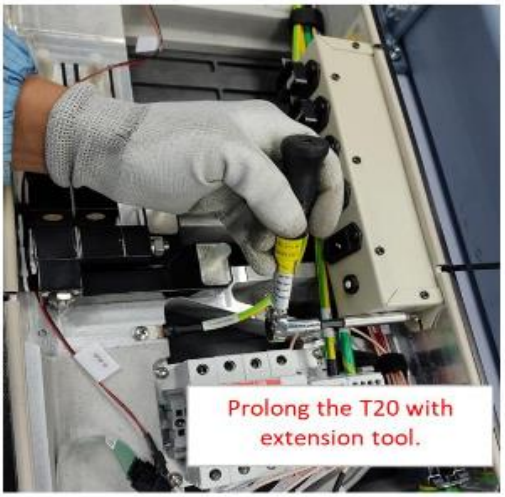
MACHINE: T20 torx wrench 1.6 N.m

METHOD:

8.5 Insert the power strip into the right screw as shown in the figure, and fix it with 1 screw (202-591-10) on the left.
Use the tool T20 torx wrench. The torque value is 1.6 N·m.
When installing the power strip, make sure that there is no wire between the power strip and the T/H frame.

8.6 Place the ground wire outside the patch panel as shown in the figure, making sure that all other wires are under the patch panel.





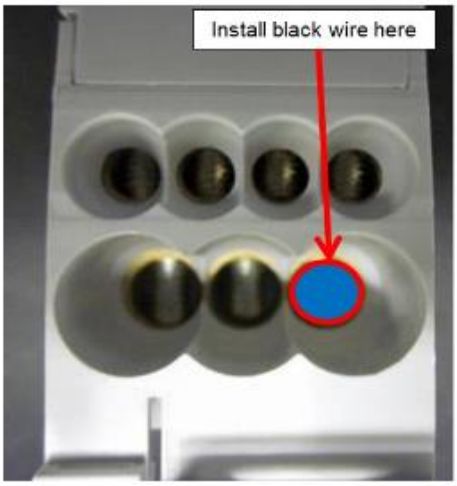
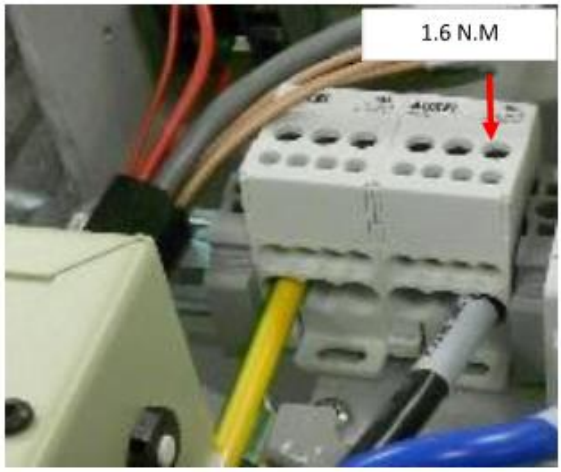
MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
42	652-436-00	CABLE UF2 BUS TO CB 48V SOLID PIN TELECOM WITH CONN	1

MACHINE: PH2 Phillips tool bit 1.6 N.m

METHOD:

9.1 Take the wire (652-436-00) as shown in the figure.

9.2 Insert the black end line "GND BLOCK" into the rightmost position of the lower layer of the power distribution block, and keep it flush with the edge of the power distribution block.
Use the PH2 Phillips tool bit with a torque value of 1.6 N·m.